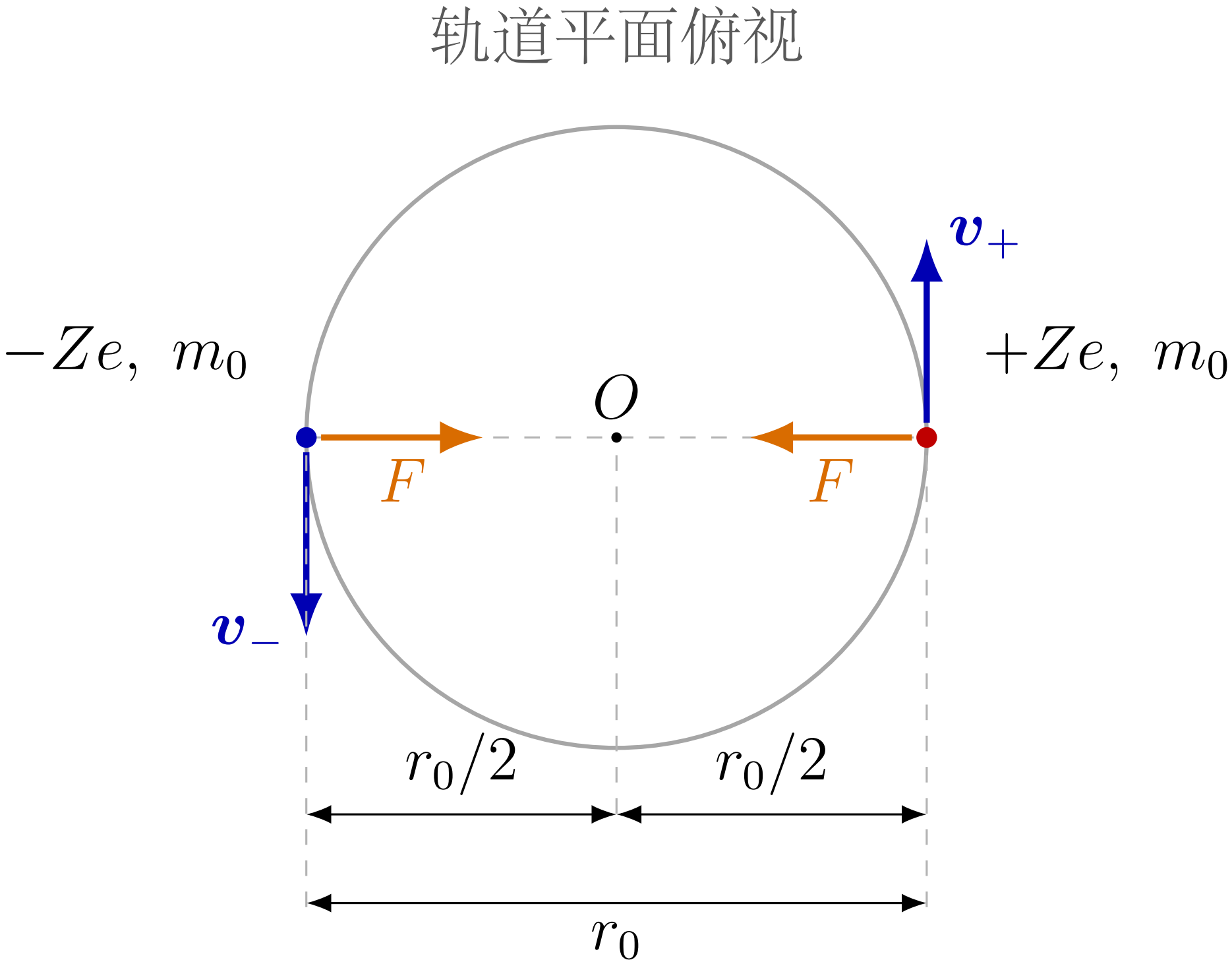
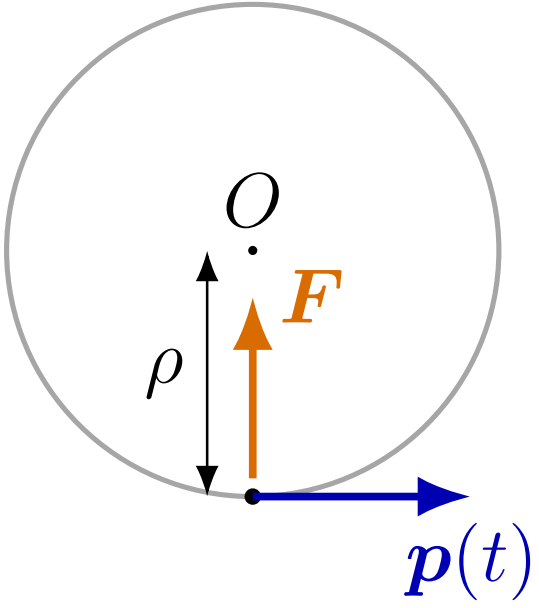
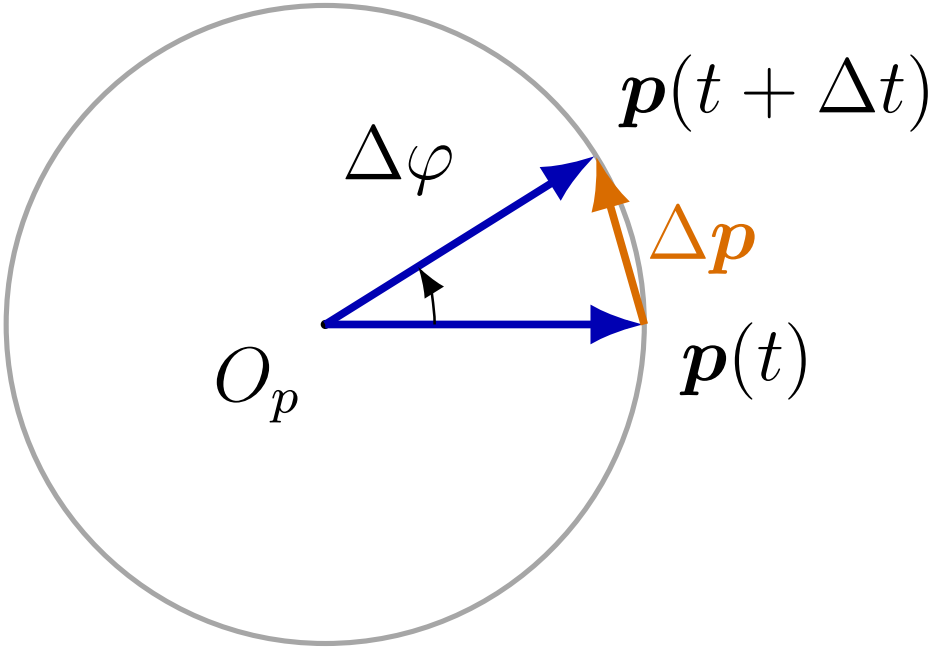

轨道平面俯视
$-Ze,\ m_0$
$+Ze,\ m_0$
O
F
F
$\boldsymbol{v}_+$
$\boldsymbol{v}_-$
$r_0/2$
$r_0/2$
r_0

位置空间

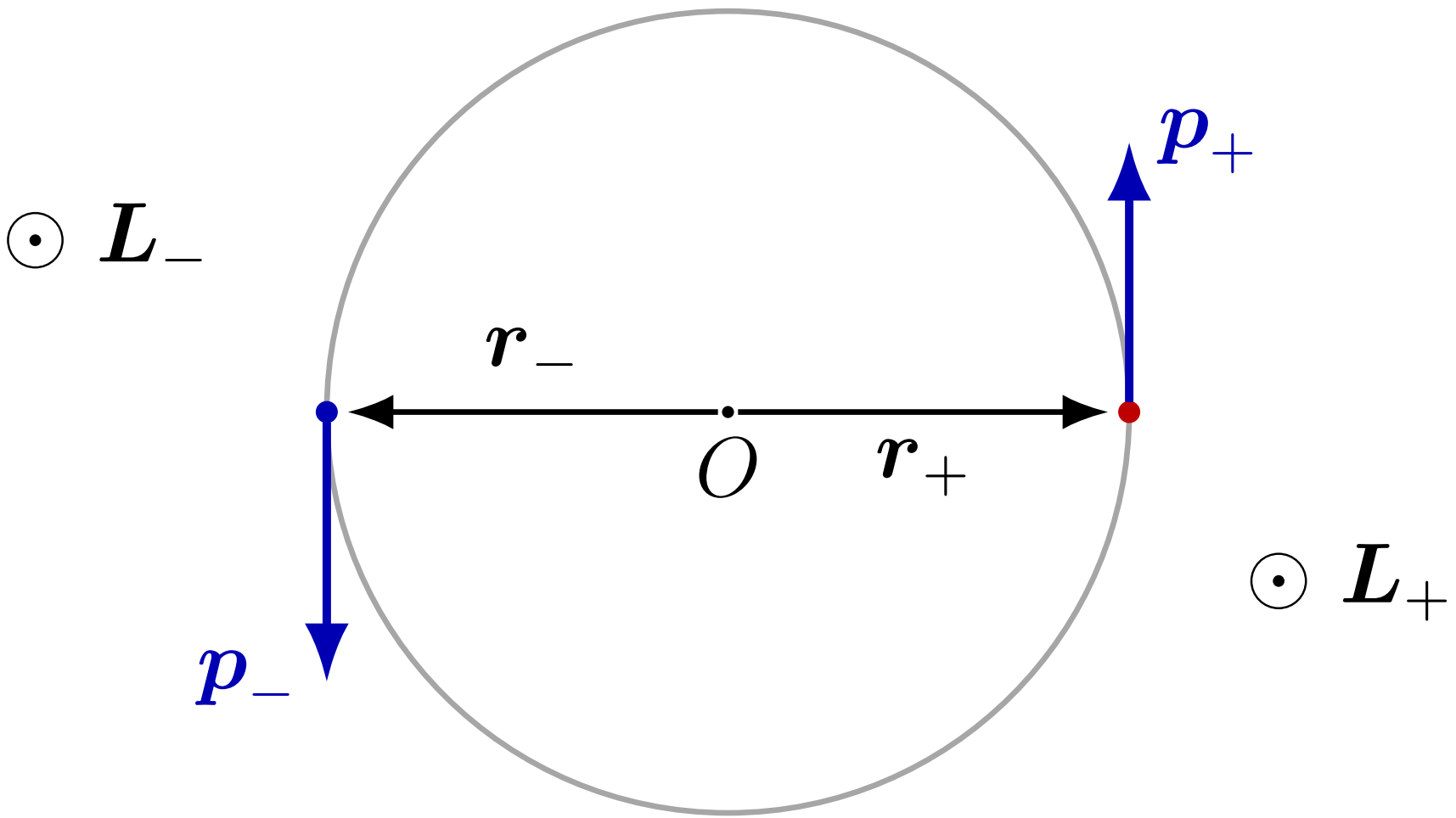


动量空间

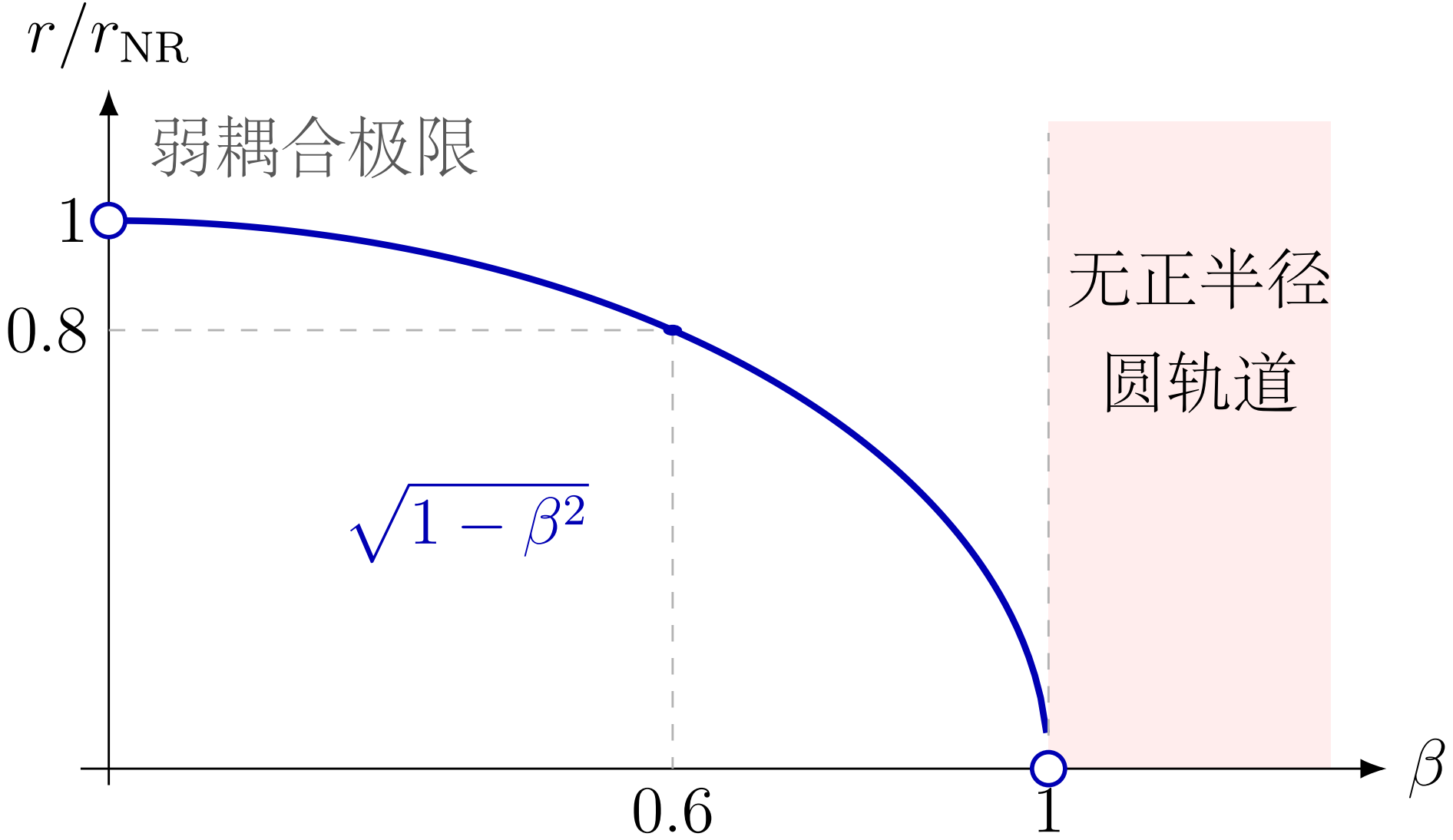


$$|\Delta\boldsymbol{p}| = 2p\sin(\Delta\varphi/2)$$

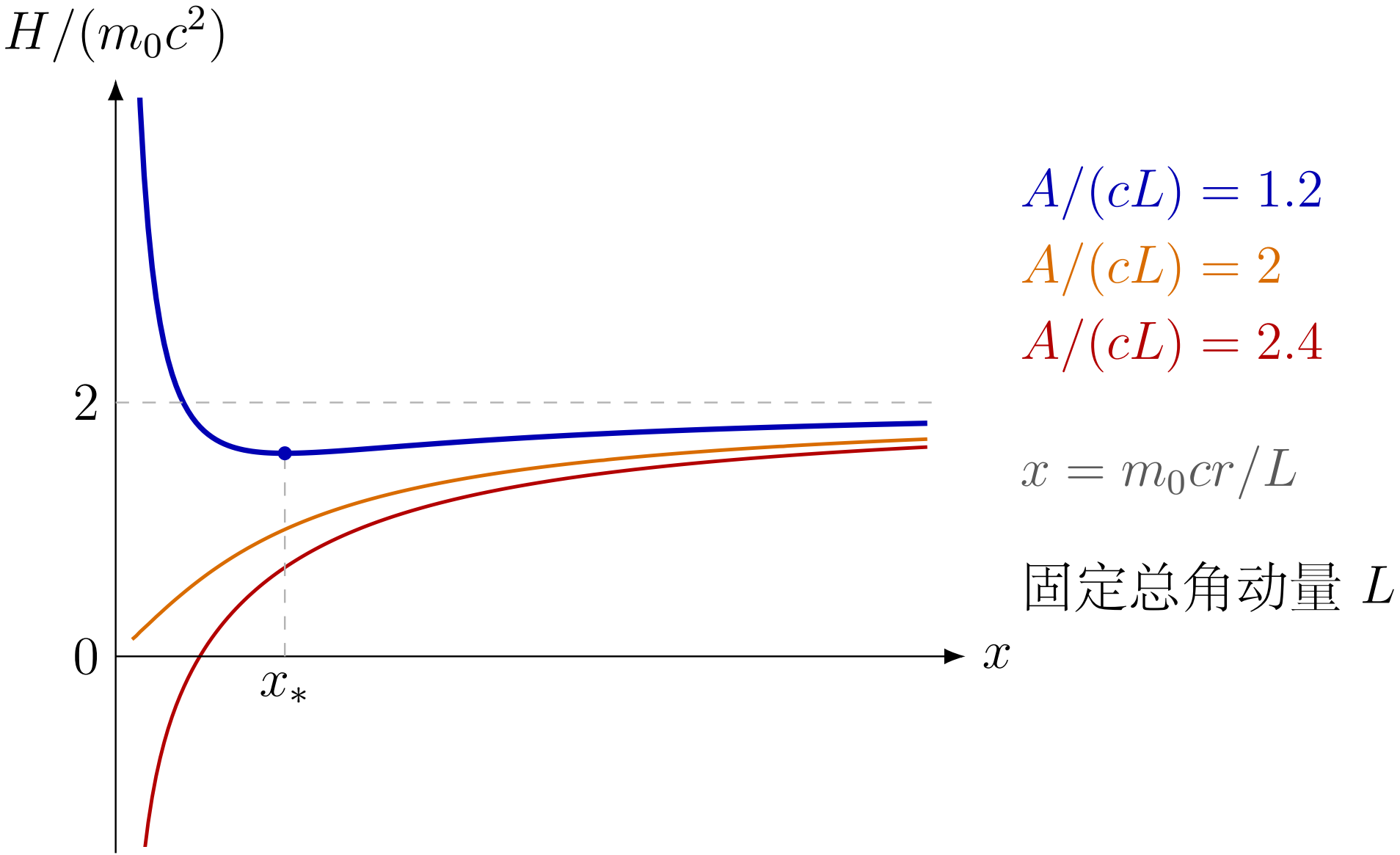
$$\boldsymbol{L} = \boldsymbol{L}_+ + \boldsymbol{L}_-$$



$$L = 2(r/2)p = rp$$



$$\beta = \alpha Z^2/(2n), \qquad r_{\mathrm{NR}} = 2n^2\hbar^2/(m_0 A)$$


$H/(m_0c^2)$
2
0
x_*
x
$A/(cL) = 1.2$
$A/(cL) = 2$
$A/(cL) = 2.4$
$x = m_0cr/L$
固定总角动量 L